

Assignment No 7

No SQL : Design & Develop queries using CRUD operations

1. CRUD OPERATIONS

CRUD operations are the foundation of database interaction.

1. CREATE Operation

Definition:

Insertion of new data into the database.

Internal Working:

- Data is converted into document format
- Stored in collection
- Unique ID (_id) is generated automatically

Example:

```
db.students.insertOne({  
  name: "Vijay",  
  age: 20,  
  course: "B.Tech"  
})
```

2. READ Operation

Definition:

Retrieving data from database based on conditions.

Internal Working:

- Query parser reads condition
- Searches indexes
- Returns matching documents

Example:

```
db.students.find({age: 20})
```

3. UPDATE Operation

Definition:

Modify existing records.

Types:

- updateOne()
- updateMany()

Internal Working:

- Finds matching document
- Applies modification using operators like \$set

Example:

```
db.students.updateOne(  
  {age: 20},  
  {name: "Vijay"},  
  {upsert: true}
```

```
{name: "Vijay"},
{$set: {age: 21}}
)
```

4. DELETE Operation

Definition:

Removes data from database.

Example:

```
db.students.deleteOne({name: "Vijay"})
```

2. SAVE METHOD (IN DEPTH)

Definition:

save() is a hybrid operation:

- Inserts new document
- Updates existing document

Working Logic:

Condition Action

_id exists

Update

_id not exists Insert

Example:

```
db.students.save({
  _id: 1,
  name: "Vijay",
  age: 20
})
```

If _id = 1 exists → update

Else → insert

Important Note:

- Deprecated in newer MongoDB versions
- Replaced by:
 - insertOne()
 - updateOne()

3. OPERATORS IN NoSQL

1. Comparison Operators

Used to compare values:

```
db.students.find({age: {$gt: 18}})
```

Operator Meaning

\$eq - Equal
 \$ne - Not Equal
 \$gt - Greater Than
 \$lt - Less Than

2. Logical Operators

\$and

All conditions must be true

```
db.students.find({
  $and: [
    {age: {$gt: 18}},
    {course: "B.Tech"}
  ]
})
```

\$or

At least one condition true

```
db.students.find({
  $or: [
    {age: 20},
    {course: "MBA"}
  ]
})
```

\$not

Negates condition

\$nor

None of the conditions true

4. JOINS IN NoSQL

NoSQL does not use traditional joins like SQL.

1. Embedding (Preferred Method)

- Data stored together in one document
- Faster retrieval

Example:

```
{
  "name": "Vijay",
  "orders": [
```

```
{ "item": "Book" },  
{ "item": "Pen" }  
]  
}
```

2. \$lookup (MongoDB Join)

Definition:

Performs join operation between collections.

Example:

```
db.orders.aggregate([  
  {  
    $lookup: {  
      from: "customers",  
      localField: "cust_id",  
      foreignField: "_id",  
      as: "customer_data"  
    }  
  }  
])
```